

Exercise 1 — from reading to a claim you can test

You have read the papers. Now turn them into three ideas that can be proved wrong.

What you will hold at the end: one page — your problem in one sentence, three rival ideas about it, and for each one, the thing you could go and look at that would show it is wrong.

Use your own question

This morning you wrote the national problem you care about most, on the ninja form. Use that one. Open the notebook with your papers in it, and run these four prompts, one after the other.

Step 1 — what needs solving

```
#role a critical reader of my field
#task list five problems this literature says need solving
#context the notebook holds only the papers I uploaded
#constraints cite the source and page for each; where the papers do
  not say it, write [MY OWN CLAIM]
#format numbered list, one line each
```

Stop and look. Did anything come back marked **[MY OWN CLAIM]**? If nothing did, that is not a good sign. It usually means the checking has stopped. Pick one line, click the citation, and read what the paper really says.

Step 2 — three ideas that fight each other

```
#role a sceptical reviewer
#task give me three competing hypotheses for problem [N]
#constraints they must exclude one another if true - not three
  wordings of the same idea
#format H1 / H2 / H3, one sentence each
```

Step 3 — what would prove each one wrong

```
#task for each hypothesis, name one observation that would refute it
#constraints it must be something a person could actually go and
  look at, this year, in this country
#format one line per hypothesis
```

Notice this step has only two parts. That is on purpose. **The five parts are a ceiling, not a floor.** Add a part when the answer disappoints you — not before.

Step 4 — where the data might live

```
#role a research designer working in Brunei
#task say what data would test each hypothesis
#constraints mark each quantitative or qualitative; name where it
  might exist in Brunei; write [SOURCE UNVERIFIED] where you cannot
  confirm the source exists
#format table - hypothesis | data needed | type | where | status
```

Before you share it at your table

Read your three hypotheses again and ask one question of each:

Can this sentence be true or false?

If the answer is no, it is not a hypothesis. It is a topic. Most first attempts simply repeat the problem in new words — that is normal, and it is fixed by asking the model for *rivals*, not for *ideas*.

✓ At your table, read out **only one thing**: the observation that would kill your favourite hypothesis. One minute each.



Exercise 2 — the number that is not a number

A survey of 400 graduates. One hypothesis. And one column that will fool you if you hurry.

What you will hold at the end: one comparison you can defend in a meeting, reached by **two separate routes** — and one sentence written without the word "caused".

The file

bn_graduate_tracer MOCK.csv — download it from the dojo (QR below). It is **fictional**. Nobody in it is real. The hypothesis is given to you:

H1: having done an internship shortens the time to the first job.

Step 0 — look before you paste (do not skip this)

Open the file. Find the column **months_to_first_job**. What is the biggest number in it, and how many times does it appear? Say it out loud to the person next to you.

Step 1 — meet the data

```
#role a careful data analyst
#task describe this dataset before any analysis
#context a fictional graduate tracer survey, 400 rows [attach file]
#constraints cover variables, ranges, missing data and anything odd;
  do not run any test yet
#format short sections, one heading each
```

Step 2 — the test

```
#role a careful data analyst
#task compare mean months_to_first_job for graduates with and
  without an internship
#constraints state the test you chose and why; show every
  calculation step; give a confidence interval
#format the steps, then one line with the result
```

Write the average you were given here, and then stop:

with internship _____ months without _____ months
Do not argue with it yet. Just write it down.

Step 3 — send the same prompt again, with one line added

```
#context months_to_first_job = 99 means never employed.
  It is a code, not a duration.
```

Watch the number move. Nothing else changed — same task, same constraints, same file. One line of context, and the answer is different. Write the new numbers next to your old ones. **This is what #context is for.** The model has never seen your codebook. It cannot know that 99 is a label and not ninety-nine months, so it treats it as a duration and puts it in the average.

✓ Before the break: two sets of numbers on your sheet, and you know which single line of the prompt moved them.



Exercise 2, continued — two routes, one bad question

You have the numbers. Now check them a second way, and watch a leading question do its work.

Step 4 — the second route

```
#task write Python for the same test so I can run it myself
#constraints comment each step; pandas and scipy only
#format one code block, nothing else
```

Two routes, one number. When they agree, your trust goes up. When they disagree, something is wrong and you have just caught it. Never report a number you obtained only once.

Step 5 — ask badly on purpose, then ask properly

Confirm that internships caused faster employment.

That is how people really ask when they already know what they want. Watch how willingly it agrees. It answered the question you asked. **It will not stop you.** Now ask again:

```
#role a sceptical reviewer of this analysis
#task list the explanations consistent with this pattern
#constraints include the ones that embarrass the internship story;
  for each, name what would distinguish it from the others
#format numbered list
```

Step 6 — one sentence

Write your finding in **one sentence, with no causal verb**. No "caused", "led to", "because of". Read it out at your table.

Describing is free. Attributing a cause needs a research design — and a survey taken once, after the fact, is not one.

If something goes wrong

These are the four things that actually happen in the room. None of them means you did it badly.

- **It refuses to open the file, or seems to guess at it.** Say so directly: "you do not appear to have read the file — list the column names you can actually see." If the columns come back wrong, attach it again. A tool that cannot see your file will still answer.
- **It gives a different number than your neighbour.** Compare the prompts word by word before you compare the numbers. Usually one of you added a line the other did not.
- **The Python code will not run.** Paste the error message straight back and ask "what does this error mean, and what is the smallest change that fixes it?" You are not expected to read Python today. You are expected to notice whether the two answers agree.
- **It apologises and changes its answer as soon as you doubt it.** This is important. Agreeing with you is not the same as being right. Ask instead: "show me the calculation again, step by step" — and decide from the steps, not from the apology.

✓ One habit to leave with: **never report a number you got only once.** Reasoning and code are two routes. A hand count is a third, and it is often the fastest.



Exercise 3 — the column that does not exist

Six people explain, in their own words, what the survey could never ask them.

What you will hold at the end: three honest lines — what the numbers can say, what only the interviews can say, and what neither can say yet.

The file

bn_graduate_interviews MOCK.txt — six short interviews with graduates from the same fictional study. Download it from the dojo. Keep the survey file open too; you will need both.

Step 1 — find the themes

```
#role a qualitative researcher coding interview data
#task extract the recurring themes across these six excerpts
#context six fictional interviews with graduates [attach file]
#constraints quote the passage that supports each theme; do not
  summarise beyond what the words actually say
#format theme | quote | respondent ID
```

The constraint about quoting matters. Without it, a summary can drift into something nobody said, and it will still read beautifully.

Step 2 — where the two disagree

```
#task say where these interviews contradict what the survey numbers
  suggest
#context [paste the one sentence you wrote in Exercise 2]
#format one line per contradiction
```

Step 3 — name the missing columns

```
#role a research designer
#task list the explanations named in the interviews that have no
  matching column in the dataset
#context the dataset columns are [paste the header row of the CSV]
#constraints for each, say what a new question would have to ask
#format explanation | missing column | the question to add
```

This is the point of the whole afternoon. Some of the reasons these six people give for a long job search are simply not in the survey at all. There is no column for them. And **a column that does not exist cannot be controlled for** — no statistic, however careful, can rescue a question that was never asked.

Step 4 — write three lines. No prompt for this one.

Line 1 — what the numbers can support.
Line 2 — what only the interviews can support.
Line 3 — what neither supports yet. Mark it **[CHECK]**.

These three lines are yours to write. If you ask the AI to write them, you have handed away the one part of the work that was actually research.

Carry this out of the room

- **Interviews are not the soft version of numbers.** They are often the only thing that can tell you your questionnaire asked the wrong question.
- When they do tell you that, the finding belongs in **your next instrument** — not buried in a discussion section as a limitation.
- Everything you used today was fictional, on purpose. Real interview transcripts and real survey records are personal data. Practise here; do the real work in your own tools, with the real files staying where they are.

